EEE 431 - Computational Assignment 2

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December 10, 2025

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Introduction

In this computational assignment, we are asked to simulate different digital modulation schemes. For the first question, we will first choose an arbitrary signal and design a binary antipodal signaling scheme. We will then design an ML rule receiver and measure its probability of error using Monte Carlo simulations at different SNRs. We will also analytically calculate the error probabilities and compare our results. We will then also simulate non-equal priors and design a MAP rule receiver. We will also implement our ML rule receiver and measure their probabilies of error. We will compare all our findings for this part. We expect the MAP receiver to perform better than the ML receiver.

For the second question, we will conduct a similar experiment with a signal space of dimensionality of 2. We will find the ML receiver and calculate its union bound of error probability. We will then use Monte Carlo simulations and compare our results. We expect the union bound to provide an upper bound and we expect it to give accurate results at high SNR values. We will work with various SNR levels. We will then compare two different constellations in terms of their probabilites of error at the same SNR level. We will also compare Gray coding with other methods in terms of the bit error probability. We expect the Gray coding to outperform other mappings.

Question 1

a)

The pulse $p(t)$ was selected as a triangular function as: $p(t) = \Lambda(t-1)$.

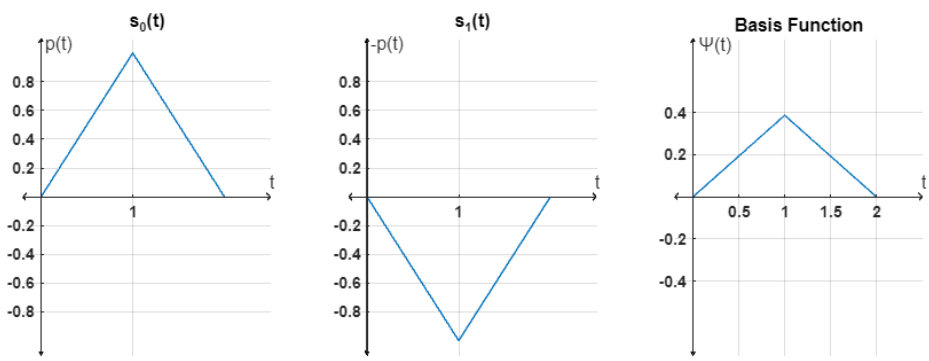

Figure 1.1.1: Signals and Basis Function of Antipodal Scheme

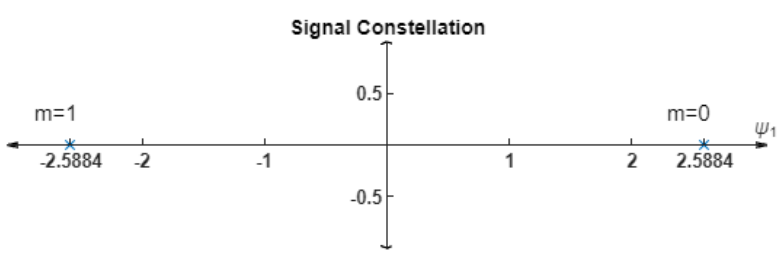

Figure 1.1.2: Signal Constellation of Antipodal Scheme

b)

For this part, we are asked to find the ML receiver, and the accompanying probability of error.

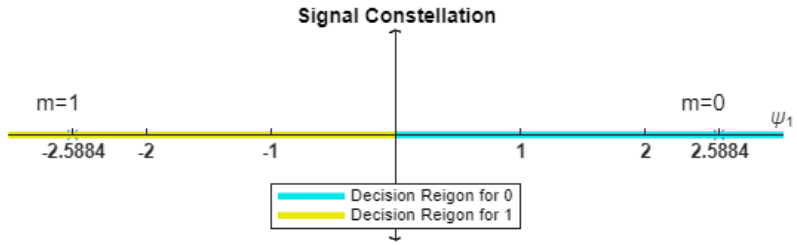

Figure 1.2.1: ML Decision Boundaries

Which would make the decision rule:

$$r \underset{1}{\overset{0}{\gtrless}} 0 \tag{1.2.1}$$

To find the probability of error, we can just use the form for binary antipodal signaling which is:

$$P(E) = Q\left(\sqrt{2\bar{\gamma}_s}\right) \tag{1.2.2}$$

c)

The sampling period T_s does not affect N_0. Because N_0 is a channel parameter that defines the power spectral density (PSD) of the noise process. Even when we reduce it to a single Gaussian random variable in the signal space representation, T_s does not change the distribution of n.

d)

When we simulate the receiver with 10 different uniformly spaced $\bar{\gamma}_s$ values ranging from 0 to 13.8311. This results in the probability of error ranging from 0.5 to 10^{-4}.

$\bar{\gamma}_s = 0$ means that the noise is infinitely large. Therefore the noise is equally likely for all values. This results in the probability that it is either greater or smaller than any finite value being 0.5.

e)

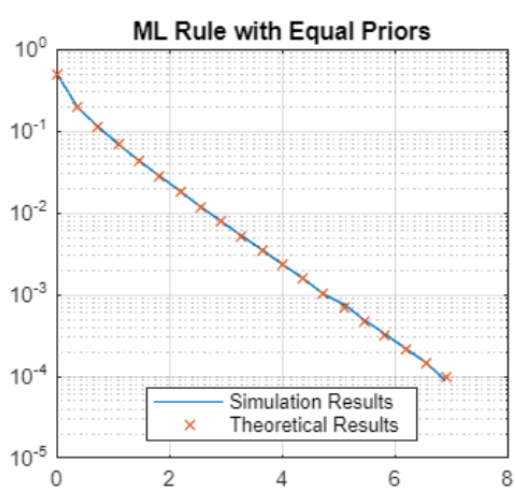

Figure 1.5.1: Probability of Error as a Function of SNR for the ML Receiver

We can see that the simulation results align nicely with the expected theoretical results.

f)

The ML receiver is not the optimal receiver when the priors are not optimal. We should select 0 when $P(0)f_r(r|0) > P(1)f_r(r|1)$ and 1 otherwise.

$$f_r(r|0) \sim N\left(E_b, \frac{N_0}{2}\right), f_r(r|1) \sim N\left(-E_b, \frac{N_0}{2}\right) \tag{1.6.1}$$

$$P(0)f_r(r|0) = p_0 \frac{1}{\sqrt{2\pi}} e^{-\frac{(r-\sqrt{E_b})^2}{N_0}} \quad \text{and} \quad P(1)f_r(r|1) = p_1 \frac{1}{\sqrt{2\pi}} e^{-\frac{(r+\sqrt{E_b})^2}{N_0}} \tag{1.6.2}$$

$$p_1 \frac{1}{\sqrt{2\pi}} e^{-\frac{(r+\sqrt{E_b})^2}{N_0}} \underset{0}{\overset{1}{\gtrless}} p_0 \frac{1}{\sqrt{2\pi}} e^{-\frac{(r-\sqrt{E_b})^2}{N_0}} \tag{1.6.3}$$

$$\frac{p_1}{p_0} e^{-\frac{(r+\sqrt{E_b})^2}{N_0}} \underset{0}{\overset{1}{\gtrless}} e^{-\frac{(r-\sqrt{E_b})^2}{N_0}} \tag{1.6.4}$$

$$\ln\left(\frac{p_1}{p_0}\right) - \frac{\left(r+\sqrt{E_b}\right)^2}{N_0} \underset{0}{\overset{1}{\gtrless}} -\frac{\left(r-\sqrt{E_b}\right)^2}{N_0} \tag{1.6.5}$$

$$\ln\left(\frac{p_1}{p_0}\right) - \frac{r^2 + 2\sqrt{E_b}r + \sqrt{E_b}^2}{N_0} \underset{0}{\overset{1}{\gtrless}} -\frac{r^2 - 2\sqrt{E_b}r + \sqrt{E_b}^2}{N_0} \tag{1.6.6}$$

$$\ln\left(\frac{p_1}{p_0}\right) \underset{0}{\overset{1}{\gtrless}} \frac{4\sqrt{E_b}r}{N_0} \Rightarrow r \underset{1}{\overset{0}{\gtrless}} \ln\left(\frac{p_1}{p_0}\right) \frac{N_0}{4\sqrt{E_b}} \tag{1.6.7}$$

The threshold for an MAP receiver along with the decision rule are given as:

$$\eta = \frac{N_0}{4\sqrt{E_s}} \ln\left(\frac{p_1}{p_0}\right), r \underset{1}{\overset{0}{\gtrless}} \eta \tag{1.6.8}$$

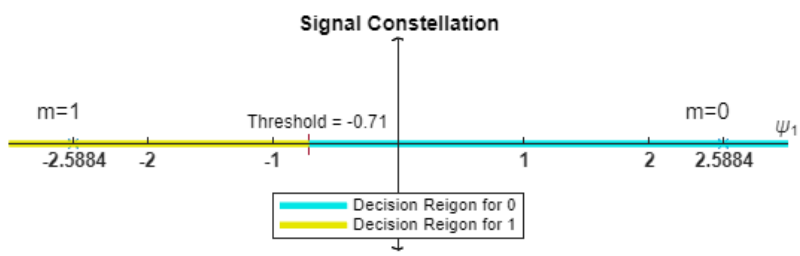

Figure 1.6.1: MAP Decision Boundaries

So the probability of error can be found as:

$$P(E) = P(S_1)P(E|S_1) + P(S_0)P(E|S_0) \tag{1.6.9}$$

$$P(E) = 0.25P(r > \eta|S_1) + 0.75P(r < \eta|S_0) \tag{1.6.10}$$

$$r|S_1 \sim N\left(-\sqrt{E_s}, \frac{N_0}{2}\right), r|S_0 \sim N\left(\sqrt{E_s}, \frac{N_0}{2}\right) \tag{1.6.11}$$

$$P(E) = 0.25Q\left(\frac{\sqrt{E_s}+\eta}{\sqrt{0.5N_0}}\right) + 0.75Q\left(\frac{\sqrt{E_s}-\eta}{\sqrt{0.5N_0}}\right) \tag{1.6.12}$$

$$P(E) = 0.25Q\left(\sqrt{2\bar{\gamma}_s} + \frac{\eta}{\sqrt{0.5N_0}}\right) + 0.75Q\left(\sqrt{2\bar{\gamma}_s} - \frac{\eta}{\sqrt{0.5N_0}}\right) \tag{1.6.13}$$

g)

Again we simulated for the same range of $\bar{\gamma}_s$ to be able to compare the results accurately.

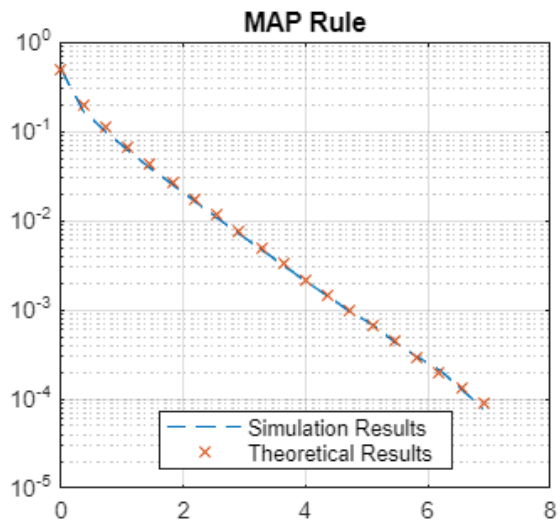

Figure 1.7.1: Probability of Error as a Function of SNR for the MAP Receiver

h)

The error of probability for the ML reciever under nonequal priors is:

$$P(E) = P(S_1)P(E|S_1) + P(S_0)P(E|S_0) = 0.25Q\left(\sqrt{2\bar{\gamma}_s}\right) + 0.75Q\left(\sqrt{2\bar{\gamma}_s}\right) = Q\left(\sqrt{2\bar{\gamma}_s}\right) \quad (1.8.1)$$

Which is the same as the receiver in part b! We can plot all of the errors together as:

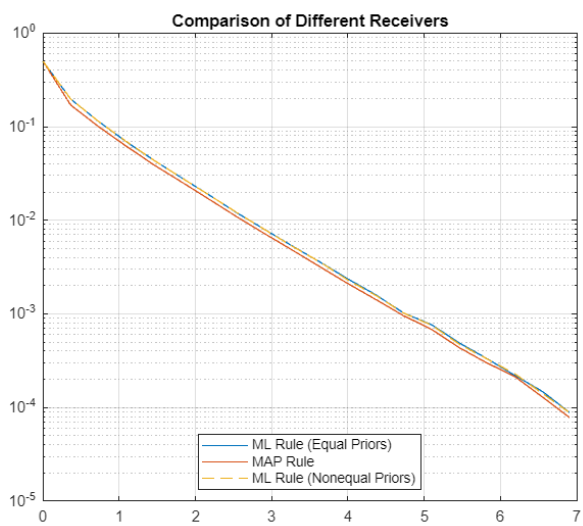

Figure 1.8.1: ML and MAP Receivers for Nonequal and Equal Priors

The ML receivers are equal as theoretically expected and since MAP receiver is the optimal receiver we can see its probability of error is smaller

Question 2

a)

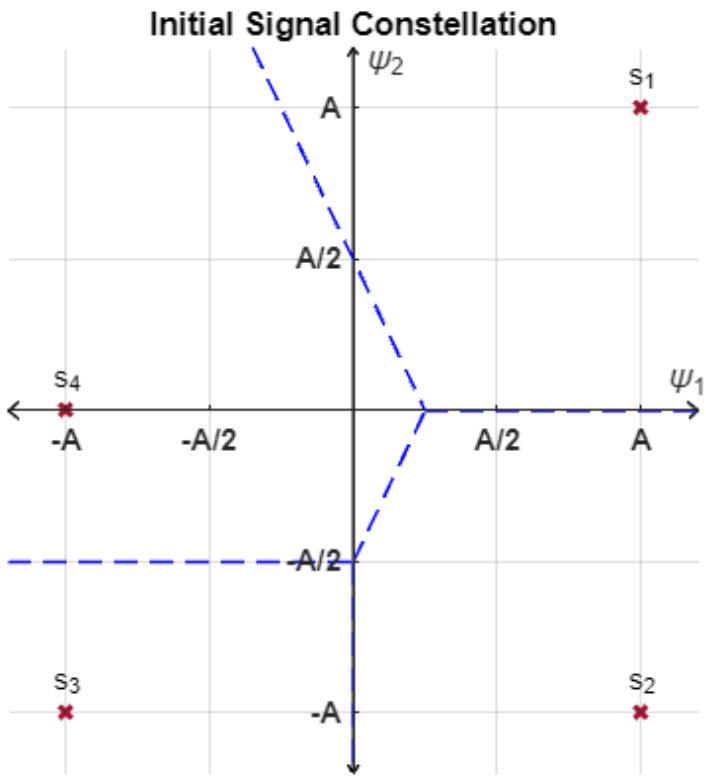

Figure 2.1.1: Signal Constellation Along with the ML Decision Rule

The union bound can be derived as:

$$P(E) = \frac{1}{4}\sum_{i=1}^{6} 2Q\left(\frac{d_1}{\sqrt{2N_0}}\right) \tag{2.1.1}$$

$$d_{12}, d_{23} = 2A, d_{14} = d_{24} = \sqrt{5}A, d_{13} = 2\sqrt{2}A, d_{34} = A \tag{2.1.2}$$

$$E_s = \frac{2A^2 + 2A^2 + A^2 + 2A^2}{4} = \frac{7}{4}A^2 \Rightarrow A = \sqrt{\frac{4E_s}{7}} \tag{2.1.3}$$

$$P(E) = Q\left(\sqrt{\frac{8E_s}{7N_0}}\right) + Q\left(\sqrt{\frac{10E_s}{7N_0}}\right) + \frac{1}{2}Q\left(\sqrt{\frac{16E_s}{7N_0}}\right) + \frac{1}{2}Q\left(\sqrt{\frac{2E_s}{7N_0}}\right) \tag{2.1.4}$$

b)

A similar algorithm to the one in Question 1 was used to simulate the error at different SNR values.

c)

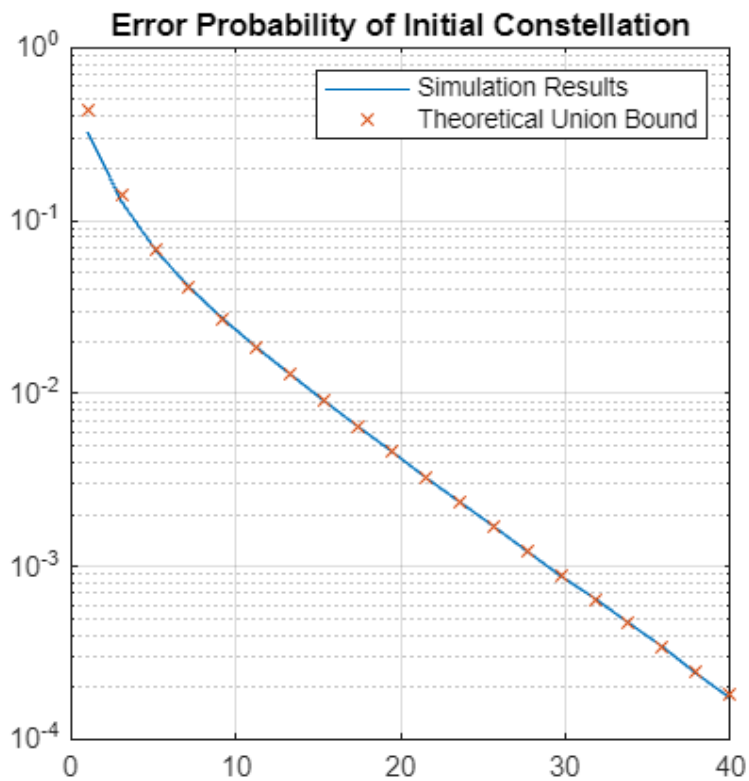

Figure 2.3.1: Union Bound and Simulation Results

For low SNR values, the union bound's accuracy decreases hence the difference between the results for low $\bar{\gamma}_s$ values. However, on the other hand, the results of union bound are highly accurate for high SNR values so our plots aligned nicely as $\bar{\gamma}_s$ increased.

d)

Similarly, we can find the constellation and decision boundaries as:

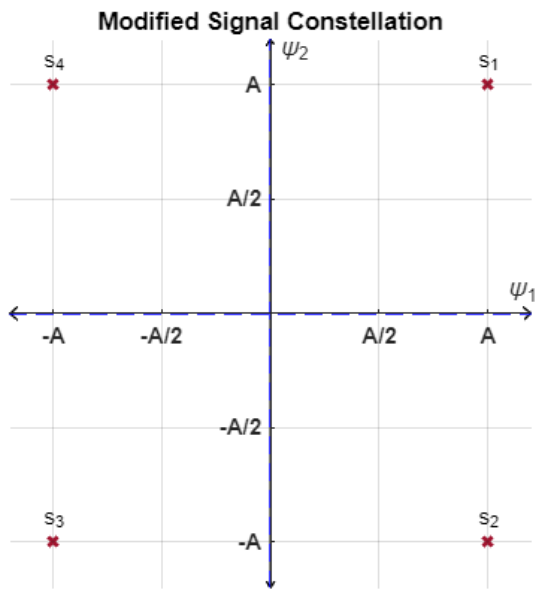

Figure 2.4.1: Updated Signal Constellation and the New Decision Reigons

This constellation will perform better since its energy is higher and consequently the distances between points are increased.

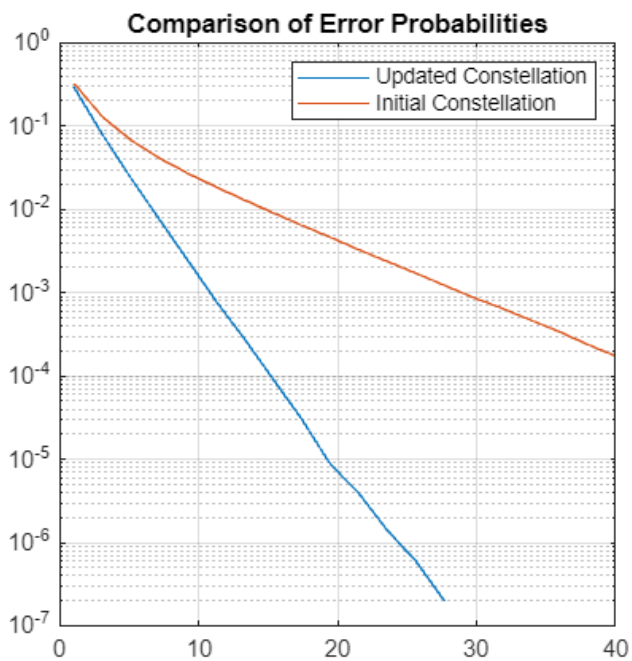

Figure 2.4.2: Updated Signal Constellation and the New Decision Reigons

Our expectation is confirmed that the updated constellation is indeed superior. The error probability is much less than the previous constellation's error probability at same the same SNR.

e)

The relation defining $\bar{\gamma}_b$ is: $\bar{\gamma}_b = 0.5\bar{\gamma}_s$. We assigned the bits to the symbols as $s_1 \Rightarrow 00, s_2 \Rightarrow 01$ $s_3 \Rightarrow 10, s_4 \Rightarrow 11$. Computing the bit error probability and comparing it with the symbol error probability we obtain:

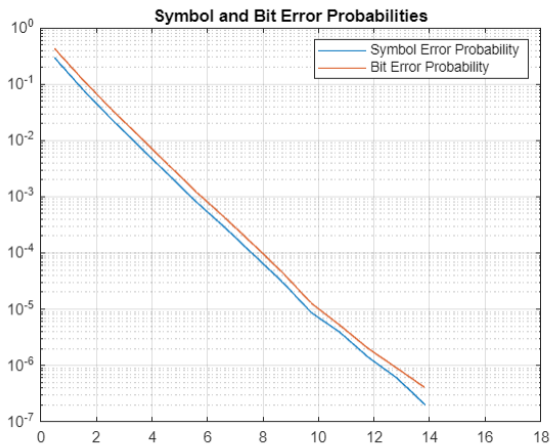

Figure 2.5.1: Bit Error vs Symbol Error Probabilites as a Function of SNR

We can see that the bit error is greater than the symbol error probability. This is because every symbol error is at least one bit error but some symbol errors can cause more than one bit error. Therefore, the bit error probability must be greater or equal to the symbol error probability.

f)

In this section we used Gray coding such that adjacent symbols differ by only 1 bit. The mapping was: $s_1 \Rightarrow 00, s_2 \Rightarrow 01, s_3 \Rightarrow 11, s_4 \Rightarrow 10$. This scheme improves bit error probability since adjacent errors are the most likely symbol errors and this coding ensures that the most likely errors only cause 1 bit error while less likely symbol errors cause 2 bit errors. This was not the case in the previous coding where s_1 and s_4 were adjacent symbols but they differed by two bits, making the likely symbol error cause 2 bit errors. Bit error probability is similar to symbol error probability at high SNRs since the less likely (2 bit error) is very rare, leading to 1 bit per symbol error. If we zoom at the low SNR values, we can observe a difference between gray coding bit error probability and symbol error probability.

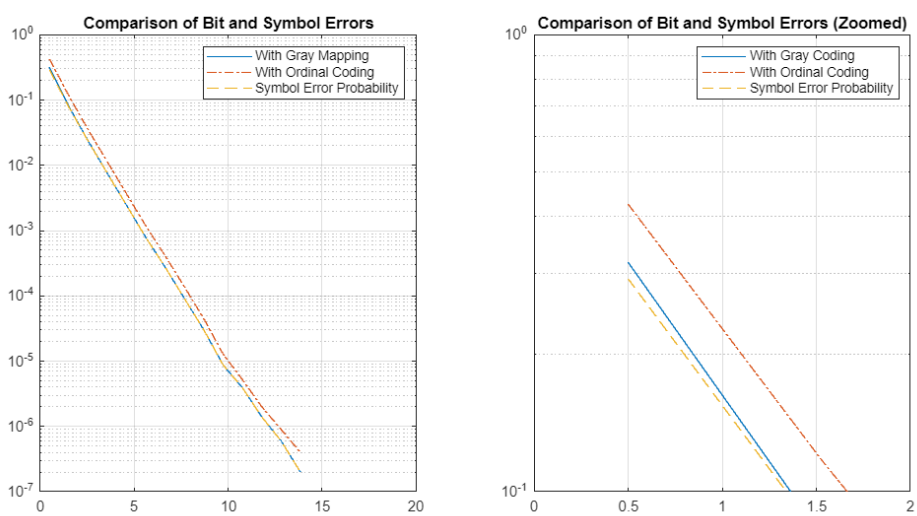

Figure 2.6.1: Bit Error vs Symbol Error Probabilites as a Function of SNR

Conclusion

In this assignment, we conducted comprehensive simulations and analytical evaluations for several digital modulation scenarios in both one–dimensional and two–dimensional signal spaces. For the binary antipodal signaling case, we verified that the simulated ML receiver performance matches the theoretical error probability expression across a broad SNR range. We also demonstrated that introducing non-equal priors requires switching to a MAP rule, and our results confirmed that the MAP receiver outperforms the ML receiver when prior probabilities are unequal, as expected.

In the second part, we extended our analysis to a two–dimensional constellation. By deriving the union bound and comparing it with Monte Carlo simulation outcomes, we observed that the bound becomes increasingly accurate at high SNR, while being looser for low SNR values. We further analyzed an alternative constellation with increased average energy, and both analytical distance calculations and simulation results confirmed its superior performance due to larger minimum distances between symbols. Finally, we compared bit and symbol error probabilities under different bit-mapping strategies and showed that Gray coding significantly improves bit error performance, especially at moderate SNRs, by minimizing multi-bit errors for nearest-neighbor symbol mistakes.

Overall, the study validated the theoretical principles of digital communication system design, including optimal detection rules, the role of constellation geometry, and the benefits of Gray coding. The simulation results consistently supported the analytical derivations, reinforcing the expected behavior of digital modulation schemes under AWGN channels.